

May 21, 2019

Dr. Heather Wood,
Chief Editor, Nature Reviews Neurology

Dear Dr. Wood,

Thank you for your positive response to my previous inquiry and providing the opportunity to submit a Consensus Paper entitled “Discovery and Validation of Biomarkers to Develop Non-Addictive Therapeutics for Pain: Challenges and Opportunities” ([NRNEUROL-19-007V1A](#)), on behalf of leaders across multiple disciplines (including clinical practice, pharmaceutical and academic research, technology developers, and patient advocacy. We have also broadened our scope as you suggested, and include initiatives in Europe and elsewhere.

As I noted in my initial correspondence with you in January, in 2017, I led a task force that led to a publication in your journal (Brain imaging tests for chronic pain: medical, legal and ethical issues and recommendations, [nature.com/articles/nrneurol.2017.122](#)). This widely read and cited paper has already been impactful with an Altmetric score “ranked 1st of the 41 tracked articles of a similar age in Nature Reviews Neurology”. The scope of this task force was focused and limited to brain imaging but in the last few years, there has been an explosion of technology and development in the biomarker field that extends well outside just brain imaging.

Thus the paper I am now submitting provides recommendations and guidelines for best practices in chronic pain biomarker discovery and validation and is the culmination of a 2 day US NIH workshop on The Discovery and Validation of Biomarkers to Develop Non-Addictive Therapeutics for Pain. This workshop convened scientific leaders across sectors including academia, industry, government, and patient advocacy groups to share and discuss progress, challenges, gaps, and ideas focused on facilitating the development of biomarkers and endpoints for pain investigations, drug development or other analgesic manipulations.

The workshop was spearheaded by US National Institutes of Health as part of the Helping to End Addiction Long-Term (HEAL) Initiative; an aggressive, trans-agency effort to speed scientific solutions to stem the national opioid public health crisis in the US. One of the research opportunity areas proposed by the Initiative is for NIH to support—in collaboration with industry leaders—biomarker discovery and rigorous validation in order to accelerate high-quality neurotherapeutic and pain clinical research.

The paper reflects discussions and topics from the workshop, provides a critical perspective on challenges and opportunities to develop pain biomarkers and endpoints, with a focus on the importance of biomarkers as tools to improve clinical practice and development of pain therapeutics, and provides recommendations. Objective biomarkers and endpoints throughout the drug discovery and development process is critical, to define pathophysiologic subsets of pain, evaluate target engagement and predict the analgesic efficacy of new drugs.

The issue of biomarkers and prognostication is a hot topic important across medicine and neurology as we move into the era of personalized and precision medicine. Thus, this paper will be of interest to the readership of Nature Reviews Neurology, for those working in the pain and a variety of fields. I look forward to your response.

Sincerely,



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